C7 Real-Time Intelligence

Eventstream, Eventhouse, KQL and Activator.

C7-L1 Eventstream, events in, routed out

Land the raw stream once, derive filtered streams for consumers, and route each to the destination that suits it.

What an eventstream does	Ingests real-time events, transforms them without code, and routes them to destinations.
Eventstream destinations	Eventhouse (direct or processed ingestion), Lakehouse (Delta tables), Activator, custom endpoint.
Permission for eventhouse, lakehouse or Activator destinations	Contributor or higher on the destination workspace.
Protect replay	Land the raw stream first. Derive filtered streams from it.

C7-L2 Eventhouse and KQL for time-series questions

Filter on time first, use top for "latest", make-series for continuous charts, and set retention and caching per table on purpose.

```
telemetry_raw
| where event_time > ago(1h)
| where event_type == "reading"
| summarize avg_rpm = avg(rpm), n = count() by vehicle_id, bin(event_time, 5m)
```

take vs top in KQL	take: arbitrary rows, not repeatable. top N by col desc: ordered, repeatable.
Gaps in time series	summarize omits empty bins. make-series ... default = 0 fills them.
KQL retention policy defaults	Default 3,650 days, maximum 36,500, minimum 1 day, or unlimited.
Caching policy	How much recent data stays in the hot SSD cache: faster queries, higher cost.

C7-L3 Activator, rules that act on events

Key each rule by the object it describes. Fire on change or sustained state, not on every event. Add a heartbeat for silence.

Events and objects in Activator	Events are grouped into objects by a shared ID. Rules run and keep state per object.
Increases above vs Is greater than	Increases above fires on the crossing. Is greater than fires on every matching event.
Detect a silent sensor	No presence of data (heartbeat) fires when no events arrive within a set time.
Activator limits to design around	10,000 events per second per rule. 30 emails or Teams messages per rule per recipient per hour.